Summary in Graph

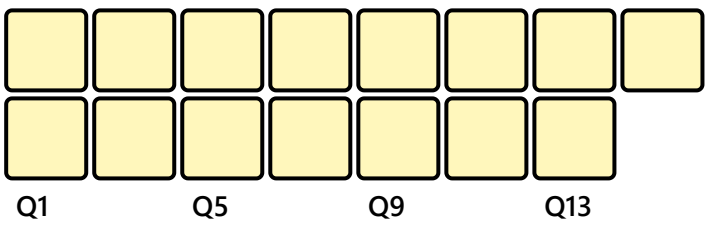
Exam Summary (GO Classes GATE CS | Computer Networks | Weekly Quiz 3)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (33) out of 332 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

In a Hamming $(7, 4)$ code (4 data bits + 3 parity bits), what is the maximum number of bit errors that can be guaranteed to be corrected?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 01sec | Discuss

Q #2 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

If one of the codewords in a Hamming $(7,4)$ code (4 data bits + 3 parity bits), is 0001011 , which of the following cannot be a valid codeword in the same group?

A. 0011101
B. 0101100
C. 0011010
D. 1110100

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #3 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

Consider a standard Hamming code where check bits are placed at positions that are powers of 2 (i.e., positions $1, 2, 4, 8, \ldots$). Suppose that bit position 13 is flipped during transmission.

Which of the following sets of check bits will detect this error by reporting incorrect parity?

A. 1, 2, and 4
B. 1, 4, and 8
C. 2, 4, and 8
D. 1, 2, and 8

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #4 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

A sender and receiver pair are using the CRC generator polynomial

$$G(x) = x^{10} + x^9 + x^5 + x^4 + x + 1$$

to protect their messages.
How long a burst error can this CRC detect for sure?

A. 9 bits
B. 10 bits
C. 11 bits
D. 12 bits

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #5 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

A coding scheme adds two parity bits to every 14-bit data block, resulting in a 16-bit codeword.

- The first parity bit covers all even-positioned bits.
- The second parity bit covers all odd-positioned bits.

What is the minimum Hamming distance between any two valid codewords in this scheme?

A. 1
B. 2
C. 3
D. 4

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

You receive the codeword 00110011111. You know that the code word includes a Hamming code for error correction, and that the check bits are computed for even parity. Bit positions increase from left to right. The spaces are added for improved readability.
Assume that the codeword either arrives correctly or contains a single-bit error.

Does received codeword contain an error? If so, what is the corrected codeword?

A. The code contains an error. The corrected codeword is 00110111111.
B. The code contains an error. The corrected codeword is 00111011111.
C. The code contains an error. The corrected codeword is 00110010111.
D. The codeword does not contain an error.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

A block of $2n$ information bits is encoded using two parity check bits:

The first parity bit covers the first n bits.

The second parity bit covers the second n bits.

These two parity bits are used for error detection at the receiver.

Which of the following statements are correct?

A. An error is detected if the number of bit errors in either half is odd.
B. If both halves contain an even number of bit errors, the error will go undetected.
C. Adding a third parity bit over all $2n$ bits ensures detection of all error patterns.
D. This scheme fails to detect errors when both halves contain an even number of bit errors.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A 4-bit message is encoded using the $(7,4)$ Hamming code (4 data bits + 3 parity bits) and transmitted over a noisy communication channel. The received 7-bit message is:

Received codeword: 0010011

The bits are arranged as follows (from left to right):

$$\mathrm{x}_7\ \mathrm{x}_6\ \mathrm{x}_5\ \mathrm{x}_4\ \mathrm{x}_3\ \mathrm{x}_2\ \mathrm{x}_1 = d_3\ d_2\ d_1\ p_4\ d_0\ p_2\ p_1$$

All parity bits use even parity. The parity conditions are:

$$\begin{aligned} p_1 &: \mathrm{x}_7 \oplus \mathrm{x}_5 \oplus \mathrm{x}_3 \oplus \mathrm{x}_1 = 0 \\ p_2 &: \mathrm{x}_7 \oplus \mathrm{x}_6 \oplus \mathrm{x}_3 \oplus \mathrm{x}_2 = 0 \\ p_4 &: \mathrm{x}_7 \oplus \mathrm{x}_6 \oplus \mathrm{x}_5 \oplus \mathrm{x}_4 = 0 \end{aligned}$$

Is there a single-bit error in the received codeword? If yes, which bit is incorrect, and what was the original 4-bit message?

A. No error; original 4-bit message is 0010

B. Error in bit position 1; original 4-bit message is 0010

C. Error in bit position 4; original 4-bit message is 0110

D. Error in bit position 5; original 4-bit message is 0011

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

The CRC of a data link layer protocol uses the following generating polynomial: $g(x) = x^{16} + x^{12} + x^5 + 1$. Which statement is correct?

A. The CRC can detect a single bit error.
B. Some 2-bit errors can be detected.
C. All errors with odd number of bits can be detected.
D. All of the above.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

The following is an error-correcting code in which any two valid codewords differ by a Hamming distance of at least 3. This allows correction of up to one-bit errors.

The encoding table and received patterns are given below:

Encoding Table **Received Patterns**

Received Pattern	Decoded Symbol
$p:\ 001111$	B
$q:\ 101000$	F
$r:\ 111110$	H
$s:\ 010010$	C

Symbol	Codeword
A	000000
B	001111
C	010011
D	011100
E	100110
F	101001
G	110101
H	111010

Which of the following mappings is correct?

A. $p \to B,\ q \to F,\ r \to H,\ s \to C$

B. $p \to F,\ q \to C,\ r \to B,\ s \to H$

C. $p \to B,\ q \to H,\ r \to F,\ s \to C$

D. $p \to H,\ q \to F,\ r \to B,\ s \to C$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A receiver uses the CRC generator polynomial

$$G(x) = x^4 + x + 1$$

and receives the bit sequence:
110110101100
After performing modulo-2 division, the remainder is found to be non-zero. What can be concluded?

A. The message was received correctly. The original data is 11011010 .
B. The message was received correctly. The original data is 110110101100.
C. The message was corrupted during transmission, and the number of bit errors is known.
D. The message was corrupted during transmission, but the number of bit errors cannot be determined.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Consider the generator polynomial $G(x) = 1101111$.
Which of the following statements is true regarding its ability to detect errors?

A. It can detect only burst errors of length 3 or less
B. It cannot detect single-bit errors
C. It can detect all errors with even number of bit flips
D. It can detect all errors involving an odd number of bit flips

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A block code uses 3 data bits D_1, D_2, D_3 and 3 parity bits P_1, P_2, P_3, resulting in a 6-bit codeword. The parity bits are calculated at the sender using equations shown in **Table A**. At the receiver, three error indicators E_1, E_2, E_3 are computed as shown in **Table B**.

Table A: Parity Bit Equations

$$\boxed{\begin{aligned} P_1 &= D_1 + D_2 \\ P_2 &= D_2 + D_3 \\ P_3 &= D_3 + D_1 \end{aligned}}$$

Table B: Error Indicator Equations

$$\boxed{\begin{aligned} E_1 &= D_1 + D_2 + P_1 \\ E_2 &= D_2 + D_3 + P_2 \\ E_3 &= D_3 + D_1 + P_3 \end{aligned}}$$

At the receiver, the computed error indicator pattern is $E_1E_2E_3 = 010$. Assuming that at most one bit error occurred during transmission, what is the most likely conclusion?

A. No errors occurred

B. A single-bit error occurred in data bit D_1

C. A single-bit error occurred in parity bit P_2

D. Multiple-bit errors occurred

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

In a communication system using CRC (Cyclic Redundancy Check), the following notations are used:

- $C(x)$: the codeword polynomial transmitted by the sender
- $R(x)$: the polynomial received by the receiver
- $G(x)$: the generator polynomial
- $E(x)$: the error pattern polynomial

Under which of the following conditions can an error go undetected?

A. When $E(x)$ has a degree less than that of $G(x)$
B. When $G(x)$ has degree less than $C(x)$
C. When the error polynomial $E(x)$ is divisible by the generator polynomial $G(x)$
D. When the received polynomial $R(x)$ is divisible by the generator polynomial $G(x)$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Consider a 12-bit Hamming codeword consisting of 8 data bits and 4 check bits arranged as:

$$d_8\ d_7\ d_6\ d_5\ c_8\ d_4\ d_3\ d_2\ c_4\ d_1\ c_2\ c_1$$

The values of the data and check bits are given below:

Data Bits

d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
0	1	1	x	1	0	1	0

Check Bits

c_8	c_4	c_2	c_1
y	1	0	1

Assume even parity is used for all check bits.

Which of the following options gives the correct values of x and y?

A. $x = 0,\ y = 0$
B. $x = 0,\ y = 1$
C. $x = 1,\ y = 0$
D. $x = 1,\ y = 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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